# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

# MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

## Mathematics (Science Group) (Tenth, 2023, Lahore Board) (Objective Type) (Group - I)

**Time Allowed:** 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Standard form of quadratic equation is:

(A) $bx + c = 0, b \neq 0$
(B) $ax^2 + bx + c = 0, a \neq 0$
(C) $ax^2 = bx, a \neq 0$
(D) $ax^2 = 0, a \neq 0$

Q2. If α, β are the roots of $7x^2 - x + 4 = 0$ then αβ is:

(A) $-1/7$
(B) $4/7$
(C) $7/4$
(D) $-4/7$

Q3. $1/\alpha + 1/\beta$ is equal to:

(A) $1/\alpha$
(B) $1/\alpha - 1/\beta$
(C) $(\alpha - \beta)/\alpha\beta$
(D) $(\alpha + \beta)/\alpha\beta$

Q4. The fourth proportional w of x : y :: v : w is:

(A) vy/x
(B) xy/v
(C) xyv
(D) x/vy

Q5. In a proportion a : b :: c : d, a and d are called:

(A) Means
(B) Fourth proportional
(C) Extremes
(D) Third proportional

Q6. A fraction in which the degree of the numerator is greater or equal to the degree of denominator is called:

(A) Proper fraction
(B) Equation
(C) Identity
(D) Improper fraction

Q7. Power set of an empty set is:

(A) φ
(B) {φ}
(C) {a}
(D) {φ, {a}}

Q8. If A ⊆ B then A ∩ B is equal to:

(A) A
(B) B
(C) φ
(D) {φ}

Q9. Mean is affected by change in --------:

(A) Value
(B) Ratio
(C) Origin
(D) Place

Q10. A data in the form of frequency distribution is called:

(A) Ungrouped data
(B) Range
(C) Histogram
(D) Grouped data

Q11. 20° = ---------

(A) 1200°
(B) 360°
(C) 630°
(D) 3600°

Q12. Through how many non-collinear points can a circle pass:

(A) One
(B) Two
(C) Three
(D) Four

Q13. Tangents drawn at the ends of diameter of a circle are ------ to each other.

(A) Perpendicular
(B) Non parallel
(C) Collinear
(D) Parallel

Q14. The length of a chord and the radial segment of a circle are congruent, the central angle made by the chord will be:

(A) 30°
(B) 45°
(C) 60°
(D) 75°

Q15. The circumference of a circle is called:

(A) Boundary
(B) Segment
(C) Tangent
(D) Chord

# Mathematics (Science Group) (Tenth, 2023, Lahore Board) (Essay Type) (Group - I)

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

QQ.2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define quadratic equation.

(ii) Write in standard form $(x+4)/(x-2) - (x-2)/x + 4 = 0$

(iii) Solve: $(2x + 1/2)^2 = 9/4$

(iv) Find discriminant: $2x^2 - 7x + 1 = 0$

(v) Evaluate: $(1 - \omega - \omega^2)^7$

(vi) Without solving, find the sum and the product of the roots of equation: $3x^2 + 7x - 11 = 0$

(vii) Find a, if the ratios $a + 3 : 7 + a$ and $4 : 5$ are equal.

(viii) $a \propto 1/b^2$ and $a = 3$, when $b = 4$, find a when $b = 8$.

(ix) Find the third proportional to $a^3$, $3a^2$.

QQ.3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define a rational fraction.

(ii) Convert the following improper fraction into proper fraction $(6x^3 + 5x^2 - 6)/(2x^2 - x - 1)$

(iii) If $A = \{2, 3, 5, 7\}$, $B = \{3, 5, 8\}$ then find $A \cup B$.

(iv) If $y = \{-2, 1, 2\}$ then make two binary relations for $y \times y$.

(v) Find a and b if $(a - 4, b - 2) = (2, 1)$

(vi) Define a subset and give one example.

(vii) Define standard deviation.

(viii) Find the geometric mean of the observation 2, 4, 8 by using basic formula.

(ix) Write two properties of arithmetic mean.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define coterminal angles.

(ii) Convert $3\pi/4$ into degrees.

(iii) Find r when $\ell = 52cm$, $\theta = 45°$.

(iv) Prove that $(1-\sin\theta)(1+\sin\theta) = \cos^2\theta$

(v) Define obtuse angle.

(vi) Define the length of the tangent.

(vii) Define circumference of a circle.

(viii) Define an arc of a circle.

(ix) Divide any arc of any length into two equal parts.

## Mathematics (Science Group) (Tenth, 2023, Lahore Board) (PART - II) (Group - I)

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

Note: Attempt THREE questions in all. But question No. 5 is compulsory.

QQ.5(a). (Marks 4) Solve the equation $2x^4 - 11x^2 + 5 = 0$

QQ.5(b). (Marks 4) Prove that : $x^3 - y^3 = (x - y)(x - xy)(x - x^2y)$

QQ.6(a). (Marks 4) Find x in the proportion $p^2 + pq + q^2 : x :: (p^3 - q^3)/(p + q) : (p - q)^2$ [right edge of the scan is cut off]

QQ.6(b). (Marks 4) Resolve into partial fractions $(x^2 - 3x + 1)/((x - 1)^2(x - 2))$

QQ.7(a). (Marks 4) If U={1,2,3,4,.....10}, A={1,3,5,7,9} and B={1,4,7,10} then verify $A - B = A \cap B'$

QQ.7(b). (Marks 4) Calculate the variance for the data: 10,8,9,7,5,12,8,6,8,2 [last two values blurry at the right edge of the scan]

QQ.8(a). (Marks 4) Prove that: $\sec\theta - \cos\theta = \tan\theta \sin\theta$

QQ.8(b). (Marks 4) Inscribe a circle in a triangle ABC with: |AB| = 5cm, |BC| = 3cm, |CA| = 3 cm.

QQ.9. (Marks 3) Prove that two chords of a circle which are equidistant from the centre, are congruent. OR Prove that the opposite angles of any quadrilateral inscribed in a circle are supplementary.

# GROUP – II

## Mathematics (Science Group) (Tenth, 2023, Lahore Board) (Objective Type) (Group - II)

**Time Allowed:** 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. $(x^3 + 1)/((x-1)(x+2))$ is :

(A) A proper fraction
(B) An identity
(C) A constant term
(D) An improper fraction

Q2. Two tangents drawn to a circle from a point outside it are of ------ in length.

(A) Half
(B) Equal
(C) Double
(D) Triple

Q3. The spread or scatterness of observation in a data set is called:

(A) Average
(B) Dispersion
(C) Central tendency
(D) Mode

Q4. How many common tangents can be drawn for two touching circles:

(A) 1
(B) 2
(C) 3
(D) 4

Q5. Cube roots of $-1$ are:

(A) $-1, -\omega, -\omega^2$
(B) $-1, \omega, -\omega^2$
(C) $1, -\omega, -\omega^2$
(D) $-1, -\omega, \omega^2$

Q6. The number of elements in power set of {1, 2, 3} is :

(A) 4
(B) 6
(C) 8
(D) 9

Q7. Through how many non-collinear points can a circle pass:

(A) One
(B) Two
(C) Three
(D) Four

Q8. In continued proportion a : b :: b : c, $ac = b^2$ b is said to be ---- proportional between a and c.

(A) Second
(B) Third
(C) Fourth
(D) Means

Q9. An arc subtends a central angle of 40° then the corresponding chord will subtend a central angle of :

(A) 20°
(B) 40°
(C) 60°
(D) 80°

Q10. Point 0, 1, 4 lies in the quadrant:

(A) I (B) II

(C) III (D) IV

Q11. If u/v = v/w = k then:

(A) $u = wk^2$ (B) $u = vk^2$

(C) $u = w^2k$ (D) $u = v^2k$

Q12. If a : b = x : y, then alternendo property is:

(A) a/x = b/y (B) a/b = x/y

(C) (a+b)/b = (x+y)/y (D) (a−b)/x = (x−y)/y

Q13. Mean is affected by change in ---------:

(A) Value (B) Ratio

(C) Origin (D) Place

Q14. Standard form of quadratic equation is :

(A) bx + c = 0, b ≠ 0 (B) $ax^2 = 0$, a ≠ 0

(C) $ax^2 = bx$, a ≠ 0 (D) $ax^2 + bx + c = 0$, a ≠ 0

Q15. $1/(1+\sin\theta) + 1/(1-\sin\theta)$ = ---------

(A) $2\sec^2\theta$ (B) $2\cos^2\theta$

(C) $\sec^2\theta$ (D) $\cos\theta$

## Mathematics (Science Group) (Tenth, 2023, Lahore Board) (Subjective Type) (Group - II)

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

QQ.2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Solve by factorization : $x^2 - 11x = 152$

(ii) Solve the equation using quadratic formula: $6x^2 - 3 - 7x = 0$

(iii) Define radical equation.

(iv) Find the discriminant of the given quadratic equation: $9x^2 - 30x + 25 = 0$

(v) Write the quadratic equation having roots: −2, 3

(vi) Use synthetic division to find the quotient and the remainder when $(x^2 + 7x - 1) \div (x + 1)$

(vii) Find a third proportional to: $a^3 - 3a^2$

(viii) Define proportion.

(ix) Find the fourth proportional to: 8, 7, 6

QQ.3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Convert the improper fraction into proper fraction: $(6x^3 + 5x^2 - 6)/(2x^2 - x - 1)$

(ii) Define improper fraction.

(iii) If A = N and B = W, then find the value of A − B.

(iv) If X = {2, 4, 6, -----, 20} and Y = {4, 8, 12, ------, 24}, then find Y − X.

(v) If A = {0, 2, 4} then find A × A.

(vi) If L = {a, b, c} and M = {d, c, f, g}, then find two binary relations in L × M.

(vii) Define Mode.

(viii) Find the arithmetic mean by direct method 12, 14, 17, 20, 24, 29, 35, 45

(ix) For the following data, find the harmonic mean: X | 12 | 5 | 8 | 4

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

(i) How many minutes are in two right angles?

(ii) Find 'r', when $\ell = 52$ cm, $\theta = 45°$.

(iii) Convert 225° into radian.

(iv) Verify that : $\sin^2\theta/\cos\theta + \cos\theta = \sec\theta$

(v) Define zero dimension.

(vi) Define tangent to a circle.

(vii) Define circumference of a circle.

(viii) Define perimeter.

(ix) The length of the side of a regular pentagon is 5 cm. What is its perimeter?

## Mathematics (Science Group) (Tenth, 2023, Lahore Board) (PART - II) (Group - II)

**Time Allowed:** 2.10 hrs. | **Maximum Marks:** 60

Note: Attempt THREE questions in all. But question No. 9 is compulsory.

QQ.5(a). (Marks 4) Solve the equation using quadratic formula: $(x+2)/(x-1) - (4-x)/(2x) = 2\ 1/3$

QQ.5(b). (Marks 4) Prove that: $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$

QQ.6(a). (Marks 4) Using theorem of componendo-dividendo, find the value of $(s-3p)/(s+3p) + (s+3q)/(s-3q)$ if $s = 6pq/(p-q)$

QQ.6(b). (Marks 4) Resolve into partial fractions: $(6x^3 + 5x^2 - 7)/(3x^2 - 2x - 1)$

QQ.7(a). (Marks 4) If U = {1,2,3,4,------,10}, A = {1,3,5,7,9} and B = {1,4,7,10}, then prove that $B - A' = B' \cup A$

QQ.7(b). (Marks 4) Find the standard deviation "S": 9,3,8,8,9,8,9,18

QQ.8(a). (Marks 4) Verify the identity: $\sin^3\theta = \sin\theta - \sin\theta\cos^2\theta$

QQ.8(b). (Marks 4) Inscribe a circle in a equilateral triangle ABC with each side of length 5 cm.

QQ.9. (Marks 3) Prove that if two chords of a circle are congruent then they will be equidistant from the centre. OR Prove that the measure of a central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.